A12 — Operasi & pemeliharaan

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Kontrol dokumen	
ID dokumen	A12
Judul	Operasi & pemeliharaan
Versi	0.1.0 (= versi aplikasi) · commit `fd1db6c`
Tanggal	2026-10-25
Status	Draf
Klasifikasi	Internal
Pemilik	Dea Saka Kurnia Putra
Penyusun	Dibangkitkan oleh `scripts/compliance/*` (Claude Code) dari sumber kebenaran repo
Penyetuju	[KONFIRMASI] — penyetuju dokumen belum ditetapkan
Lingkup	Deploy, konfigurasi, data, insiden
Standar acuan	ISO/IEC/IEEE 12207:2017 (proses operasi & pemeliharaan)
Dokumen terkait	A09, C07

Riwayat perubahan

Versi	Tanggal	Perubahan
0.1.0	2026-10-25	Versi pertama, dibangkitkan dari sumber kebenaran repo.

1. Deploy

- `main` → Vercel produksi (https://finandex.vercel.app); cadangan Netlify.
- Rollback: Vercel → Deployments → *Promote* deployment sebelumnya; atau revert commit.
- Rilis: tag `v*` memicu `compliance.yml` (SBOM, tanda tangan, atestasi, dokumen).

2. Variabel lingkungan (tanpa nilai)

Variabel	Lingkup
`ANTHROPIC_API_KEY`	server
`ANTHROPIC_MODEL`	server
`ANTHROPIC_MODEL_FAST`	server
`ANTHROPIC_FALLBACKS`	server
`DATA_BASE_URL`	server
`CORS_ALLOWED_ORIGINS`	server
`CHAT_DEADLINE_MS`	server
`DAILY_TOKEN_BUDGET`	server

Variabel	Lingkup
`UPSTASH_REDIS_REST_URL`	server
`UPSTASH_REDIS_REST_TOKEN`	server
`LOG_LEVEL`	server
`APP_VERSION`	server
`BUILD_ID`	server
`VITE_DISTRIBUTION`	server
`CHAT_MODE`	server
`VITE_ANDROID_APK_DIRECT`	klien (publik)
`VITE_API_BASE_URL`	klien (publik)
`VITE_APP_STORE_URL`	klien (publik)
`VITE_APP_VERSION`	klien (publik)
`VITE_BUILD_ID`	klien (publik)
`VITE_DATA_BASE_URL`	klien (publik)
`VITE_DISTRIBUTION`	klien (publik)
`VITE_MS_STORE_URL`	klien (publik)
`VITE_PLAY_STORE_URL`	klien (publik)
`VITE_SUPPORT_EMAIL`	klien (publik)
`VITE_TESTFLIGHT_URL`	klien (publik)

3. Skrip npm

Skrip	Perintah
`dev`	`vite`
`build`	`npm run build:api && vite build && npm run typecheck:server`
`build:ci`	`npm run data:fetch && npm run data:verify && npm run data:extract && npm run data:fetch-regulations && npm run…`
`preview`	`vite preview`
`data:fetch`	`tsx scripts/data/fetch-inputs.ts`
`data:verify`	`tsx scripts/data/verify-inputs.ts`
`data:extract`	`tsx scripts/data/extract.ts`
`data:build`	`tsx scripts/data/build-all.ts`
`lint`	`eslint . --max-warnings=0`
`typecheck`	`tsc -p tsconfig.app.json --noEmit && npm run typecheck:server`
`typecheck:server`	`tsc -p tsconfig.node.json --noEmit`
`test`	`vitest run --coverage`
`test:golden`	`vitest run --project golden`
`test:tools`	`vitest run --project tools`

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Skrip	Perintah
`test:parity`	`tsx scripts/data/parity.ts`
`eval:chat:mock`	`tsx scripts/eval/eval-chat.ts --mock`
`eval:chat`	`tsx scripts/eval/eval-chat.ts --live`
`e2e`	`playwright test`
`a11y`	`playwright test tests/a11y`
`lighthouse`	`lhci autorun`
`size`	`size-limit`
`check:copyright`	`tsx scripts/brand/check-copyright.ts`
`check:licenses`	`tsx scripts/security/check-licenses.ts`
`check:secrets`	`tsx scripts/security/check-secrets.ts`
`verify`	`npm run lint && npm run ui:audit && npm run typecheck && npm run check:copyright && npm run test && npm run te…`
`cap:sync`	`npm run build:native && cap sync`
`cap:android`	`npm run build:native && cap sync android && cap open android`
`cap:ios`	`npm run build:native && cap sync ios && cap open ios`
`tauri:dev`	`tauri dev`
`tauri:build`	`tauri build`
`assets:generate`	`python3 -I scripts/brand/make-native-assets.py`
`icons:generate`	`tsx scripts/brand/make-icons.ts`
`assets:pdfjs`	`tsx scripts/build/copy-pdfjs-assets.ts`
`predev`	`npm run assets:pdfjs && npm run whitepaper:copy`
`prebuild`	`npm run assets:pdfjs && npm run whitepaper:copy && npm run check:licenses`
`build:native`	`cross-env VITE_API_BASE_URL=https://finandex.vercel.app VITE_DATA_BASE_URL=https://finandex.vercel.app/data VI…`
`build:api`	`tsx scripts/build/bundle-api.ts`
`whitepaper:copy`	`tsx scripts/whitepaper/copy-whitepaper.ts`
`data:fetch-regulations`	`tsx scripts/data/fetch-regulations.ts`
`data:law`	`tsx scripts/data/build-all.ts mylaiwyer finlaiwyer`
`data:antiscam`	`tsx scripts/data/build-all.ts antiscamdex`
`test:flows`	`vitest run tests/golden/kb.test.ts -t "flow engine"`
`feature-guide:check`	`vitest run tests/golden/kb.test.ts -t "feature guide"`
`ui:audit`	`tsx scripts/ui/audit.ts`
`ui:visual`	`playwright test -c playwright.visual.config.ts --update-snapshots=missing`
`agents:gen`	`tsx scripts/build/gen-agents.ts`
`eval:agents`	`tsx scripts/eval/eval-agents.ts`
`sbom:generate`	`tsx scripts/compliance/sbom.ts`

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Skrip	Perintah
`sbom:merge`	`tsx scripts/compliance/sbom.ts`
`vex:generate`	`tsx scripts/compliance/sbom.ts`
`sbom:verify`	`tsx scripts/compliance/check-sbom-min-elements.ts`
`sbom:sign`	`bash scripts/compliance/sign-sbom.sh`
`sbom:attest`	`echo "Attestations are created by .github/workflows/compliance.yml (actions/attest-sbom, actions/attest-build-…`
`sbom:docs`	`tsx scripts/compliance/build-sbom-docs.ts`
`sbom:docx`	`tsx scripts/compliance/build-sbom-docx.ts`
`docs:build`	`npm run sbom:docs && tsx scripts/compliance/build-docs.ts && npm run docs:pdf`
`docs:pdf`	`tsx scripts/compliance/pdf.ts docs/compliance/technical/A*.md docs/compliance/sbom/B*.md docs/compliance/priva…`
`docs:verify`	`tsx scripts/compliance/verify-docs.ts`
`compliance:all`	`npm run sbom:generate && npm run sbom:verify && npm run docs:build && npm run docs:verify`
`live:fetch`	`tsx scripts/live-data/fetch.ts`
`live:validate`	`tsx scripts/live-data/build.ts --dry`
`live:build`	`tsx scripts/live-data/build.ts`
`live:report`	`tsx scripts/live-data/report.ts`
`live:verify`	`tsx scripts/live-data/verify.ts`

4. Pembaruan data

`npm run data:fetch` → `data:verify` (hash) → `data:extract` → `data:fetch-regulations` → `data:build` → `test:golden`.

5. Runbook insiden

1. Deteksi (galat platform, laporan pengguna, pemindai CI).
2. Klasifikasi: insiden keamanan biasa atau **kegagalan pelindungan data pribadi** → ikuti C07 (pemberitahuan 3 × 24 jam).
3. Mitigasi: rollback deployment, rotasi secret di dashboard platform, nonaktifkan chat (`CHAT_MODE=mock`).
4. Catat di register insiden (C07 §5).

6. Pemantauan & log

Log platform Vercel berisi metadata permintaan (C01 D04); retensi [KONFIRMASI].

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